Lecture 8:
AI agents in business workflows

Business Applications (Betriebswirtschaftliche Anwendungen)

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Course content

1	Course Kickoff
2	Introduction to ERP systems
3	SAP ERP
4	Principles of ERP systems
5	~~Modern ERP concepts~~ Instead: ERP system presentations
6	Fundamentals of AI agents
7	AI agent ecosystem + economics
8	**AI agents in business workflows**
9	Agentic ERP in S/4 HANA
10	OpenClaw & Test preparation
11	E-Test

Learning objectives

At the end of the lecture, students are able to …

- … distinguish **AI agent workflows** from **RPA**
- … compare four levels of **human-AI collaboration** and match the right framework to specific business scenarios
- … identify **uniquely human domains** that resist automation
- … outline **workplace readiness** steps

Agenda

1 **Reimagining business workflows with AI agents**

2 Human-AI collaboration

3 Human skills, workplace readiness and case studies

From RPA to AI agents

The transformation **from rule-based RPA to GenAI agents** depicts a **fundamental shift in business automation**. This evolution has expanded the scope **from simple, repetitive tasks to sophisticated, context-aware operations** that can handle ambiguity and learn from experience.

"RPA is like having a very efficient but inflexible worker who can only follow exact instructions."
(anonymous industry expert, see Huang 2025, p. 135)

Skills of AI agents over RPA

Natural language understanding

Unlike RPA, which follows predetermined rules for repetitive tasks, AI agents can **interpret and act on instructions in natural language**

Contextual decision-making

Unlike RPA, which fails when encountering unexpected scenarios, AI agents **analyze context and adapt to new situations**

Continuous learning

Unlike RPA, which doesn't improve itself, AI agents **learn from interactions and feedback** and get better over time without reprogramming.

Sources: Huang (2025)

Activity: Why integrating AI agents into business workflows?

- Own brainstorming with researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)

AI agents reimagine business workflows across eight dimensions

1 From Linear to Dynamic Flows

AI agents enable **real-time adaptive workflows** – replacing rigid, predefined sequences

2 Predictive, not Reactive Processes

AI agents use ML models to **anticipate disruptions in processes** and trigger preemptive actions

3 Personalization at Scale

Real-time decision-making algorithms deliver **highly individualized user experiences**

4 Continuous Learning

Self-supervised learning with **new data and user feedback loops** refine operations autonomously

5 Breaking Department Silos

AI agents create unified data architectures to **integrate fragmented, cross-department processes**

6 Autonomous Decision-Making

Real-time analytics and rule-based governance models facilitate autonomous decisions

7 Ecosystem Integration

Distributed ledger techs, APIs and IoT networks **connect ecosystem stake-holders** (users, suppliers, ...)

8 Key Deployment Factors

Business value, transaction volume and timing (sync vs. async) guide **AI agent deployment**

Sources: Huang (2025)

AI agent workflow categorization matrix

Business Value \ Transaction Volume	Low	Moderate	High
High	Real-time trading algorithms (sync)	Crisis management (sync)	Dynamic pricing in e-commerce (sync)
Moderate	High-stakes negotiations (sync)	Moderate-scale invoice processing (async)	FAQ chatbots (sync)
Low	Monthly compliance reporting (async)	On-demand support (sync)	Archival data retrieval (sync)

Sources: Huang (2025)

AI agents enable full automation across six key areas

Data Processing

AI agents collect data from **diverse data sources** (e.g., databases, files and APIs) and **execute data cleaning, feature extraction, etc.**

Language Translation

Neural machine translation (NMT) engines translate languages while **preserving the original meaning, context and cultural nuances**

Quality Control

Computer vision and probabilistic reasoning **detect defects and analyze root causes** across data sources (e.g., engineering manuals, historical defect logs)

Customer Service

AI agents **understand intent, sentiment and even unspoken needs driving customer inquiries** leading to new service models

Predictive Maintenance

AI agents **accurately predict equipment failures before they occur** and continuously improve their own performance

Supply Chain Optimization

Self-learning, self-adapting systems **navigate global supply chain complexities** (e.g., optimize flow of goods/info across supply chain)

Sources: Huang (2025)

Full automation has five key implications for the workforce

1. **Skill Shift**
   Increasing demand for AI development, maintenance and human-AI collaboration skills

2. **Job Displacement**
   Routine cognitive roles are being eliminated, new roles are being created in AI-related fields

3. **Productivity Boost**
   Workers freed from routine tasks to focus on strategic, creative and interpersonal aspects of work

4. **Continuous Learning**
   Rapid AI development demands a culture of continuous learning and adaptation in the workforce

5. **Ethics Specialists**
   Growing need for ensuring AI systems are fair, transparent and aligned with human values

Sources: Huang (2025)

Agenda

1. Reimagining business workflows with AI agents
2. **Human-AI collaboration**
3. Human skills, workplace readiness and case studies

Four frameworks define how humans and AI agents collaborate

Agent as Assistant | Low Autonomy

Responsive helper performing tasks under **direct human guidance**. Handles drafting, scheduling, organizing. Humans maintain full control over decisions and outcomes.

Agent as Advisor | Moderate Autonomy

Sophisticated consultant providing analysis, recommendations and evidence-based insights. **Humans make final decisions using AI-powered intelligence**.

Agent as Autonomous Worker | High Autonomy

Operates with significant autonomy within defined parameters. Initiates actions, makes decisions and completes complex tasks with **minimal human intervention**.

Agent as Autonomous Org. | Full Autonomy

Self-organizing, self-managing AI systems using multiple specialized AI agents to operate business functions or even complete organizations with **minimal human intervention**.

Sources: Huang (2025)

Human-AI collaboration evolves across four levels of autonomy

Assistant	Advisor	Autonom. Worker	Autonomous Org.
Drafting, scheduling, organizing	Analysis, insights, recommendations	Routine decisions, customer service	Self-organizing, multi-agent systems
Human Control: High	**Human Control:** Medium-High	**Human Control:** Low	**Human Control:** Minimal
AI Autonomy: Low	**AI Autonomy:** Medium	**AI Autonomy:** High	**AI Autonomy:** Very High
e.g., e-mail drafting	*e.g., market analysis*	*e.g., customer inquiries*	*e.g., algorithmic trading*

Increasing AI autonomy →

← Increasing human oversight

Sources: Huang (2025)

Assistant agent drafts e-mail

Low autonomy | Human controls every step

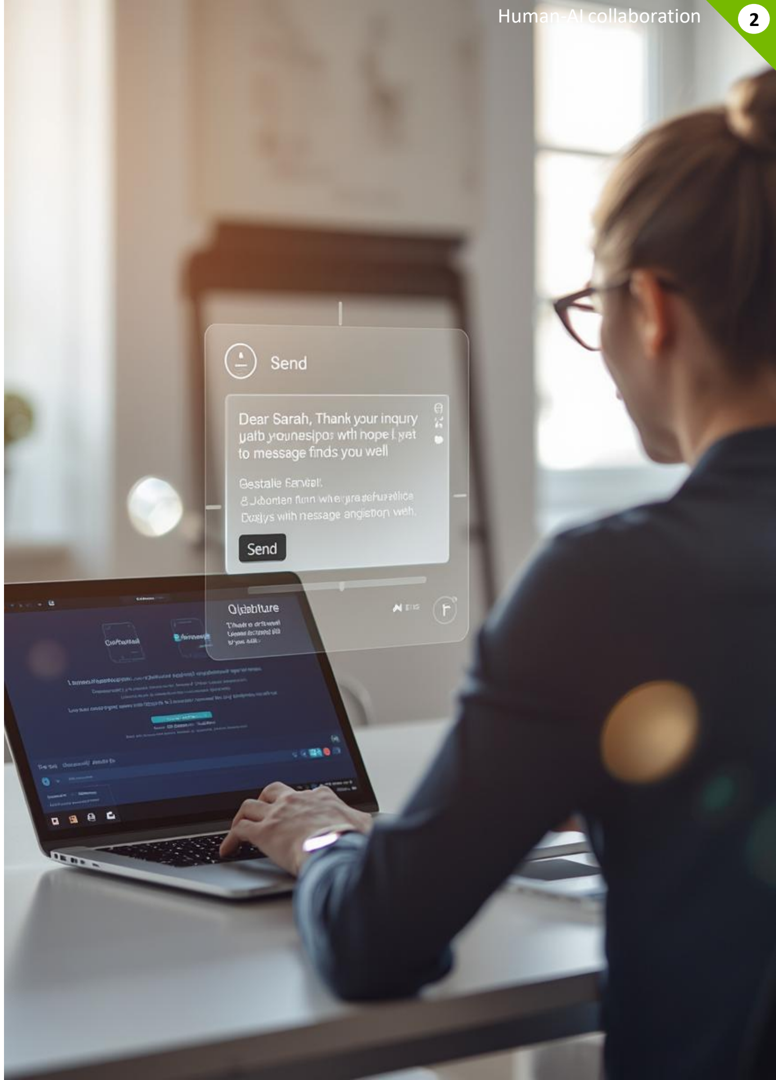
User prompts
User provides context (e.g., recipient, topic, tone)

AI drafts e-mail
Agent generates draft with subject line, greeting, body, etc.

User reviews
User reviews, edits wording and adds personal touches

User sends
Final approval and sending remains under full human control

AI saves time on drafting – human retains full editorial control



Advisor agent analyzes market

Medium autonomy | AI analyzes, human decides

User defines scope
User specifies industry, competitors, metrics, etc. to analyze

AI gathers data
Agent autonomously collects market reports, financial data, news and trend signals

AI synthesizes insights
Agent produces structured analysis with charts, SWOT and strategic recommendations

User decides
User evaluates AI recommendations and makes decisions

AI researches – human retains strategic decision authority

Autonom. worker handles inquiries

High autonomy | AI acts, human supervises

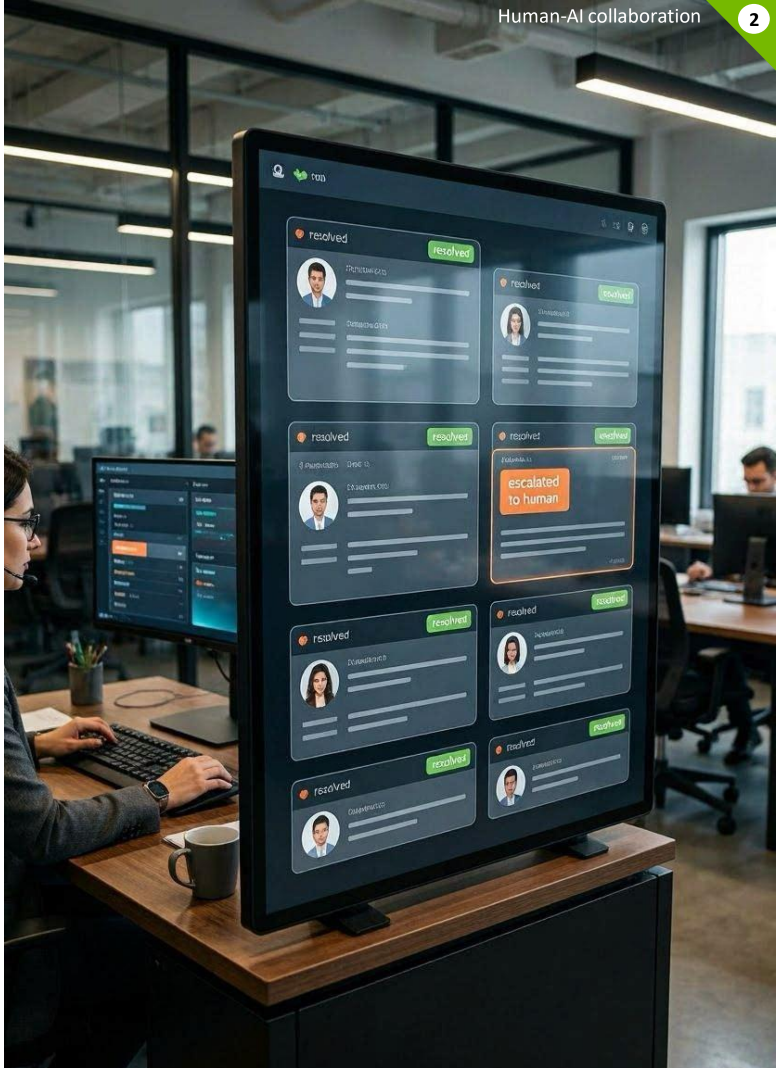
Customer reaches out
Inquiry arrives via chat, e-mail or phone – AI agent receives and classifies it automatically

AI resolves on its own
Agent accesses knowledge base, order data and company policies to solve the issue in real-time

Escalation if needed
Complex or sensitive cases are flagged and routed to a human with full context summary

AI logs and learns
Agent documents the interaction and feeds insights back to improve future resolution quality

AI handles most inquiries E2E – human handles exceptions



Autonomous org. conducts trading

Very high autonomy | AI operates, human governs

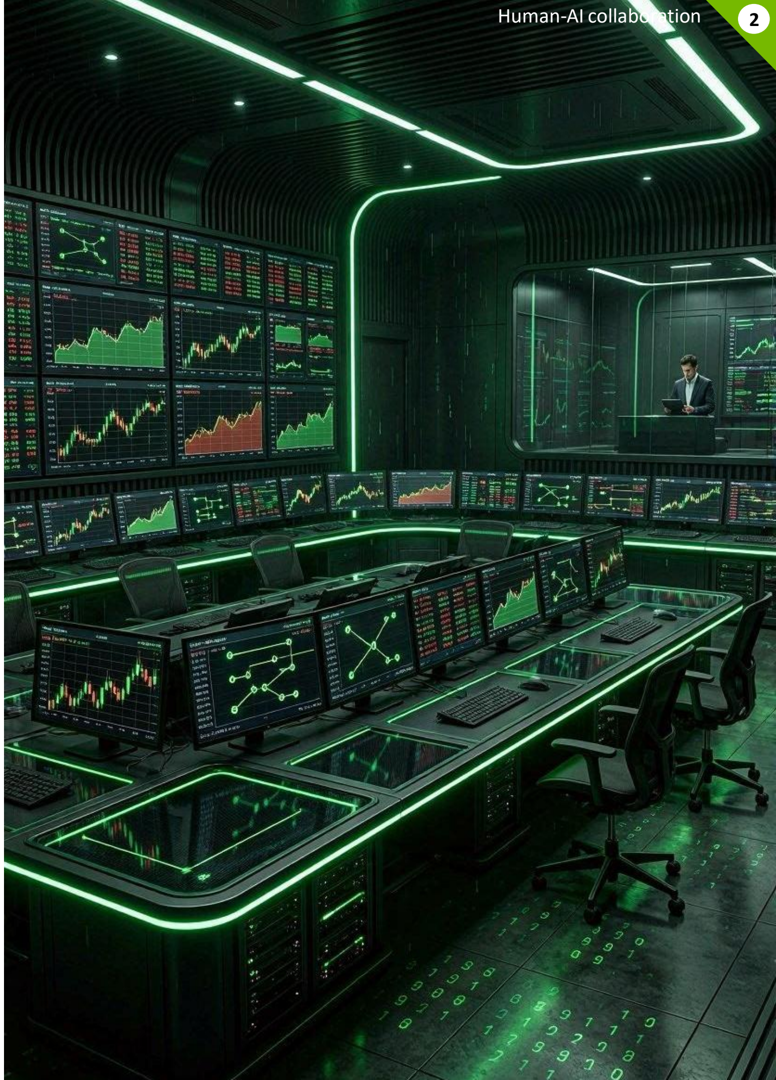
AI monitors market continuously
AI agents scan markets, news feeds, economic indicators 24/7

AI decides autonomously
AI evaluates risks, identifies opportunities and executes trades

AI coordinates agents
Specialized agents for risk, compliance, portfolio and execution collaborate as a virtual team

Human sets guardrails only
Humans define risk limits, compliance rules and strategy parameters – but never intervene in real-time

Fully autonomous AI org. – human only sets system boundaries



Practice Insight: SAP's vision of the autonomous enterprise

SAP Sapphire conference (Orlando, May 2026)

https://k2university.com/news/sap-sapphire-2026-the-autonomous-enterprise-takes-shape/

Full Keynote (1,5h)   Keynote Excerpt (10min)

Sources: SAP on YouTube

Agenda

1 Reimagining business workflows with AI agents

2 Human-AI collaboration

3 **Human skills, workplace readiness and case studies**

Activity: In which domains can’t AI agents replace humans?

- Own brainstorming without researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)

Seven domains remain where AI agents cannot replace humans

Complex Negotiations
Reading subtle emotional cues, building rapport, adjusting strategies

Strategic Leadership
Intuiting market trends, inspiring teams, making bold decisions

Creative Innovation
Drawing unexpected connections, using cultural insights for original ideas

Ethical Judgment
Moral reasoning, fairness considerations, navigating cultural sensitivities

Problem-Solving
Navigating ambiguous, unprecedented situations with creative speculation

Empathetic Relations
Deep emotional intelligence for conflict resolution and customer communication

Crisis Adaptation
Rapidly pivoting without historical precedent (e.g., COVID-19 response)

Sources: Huang (2025)

Seven steps to prepare for the AI-integrated workplace

Assess
Identify repetitive, data-driven processes with **automation** potential

Build infra.
Build scalable **data infrastructure** with governance and security

Upskill
Develop technical, analytical and problem-solving **skills**

Redesign
Define clear AI and human **roles** with escalation paths

Change
Manage change with clear **communication** and strong **leadership**

Govern
Enable **transparency**, **fairness**, **accountability** and **bias mitigation**

Evolve
Reassess **workflows**, update **systems** and invest in **innovations**

Sources: Huang (2025)

Practice Insight: Marc Benioff on AI replacing workforce

"Today's CEOs will be the last to manage all human workforces."

Marc Benioff, CEO Salesforce (June 2025)

https://www.youtube.com/shorts/YSSZHoyUynU

Sources: Bloomberg on YouTube

Case studies show AI agent impact across industries (1/2)

Olive
Healthcare

What it does
AI agents automate billing, claims and admin tasks across 675+ hospitals in 42 US states

Impact
$100M annual savings, affecting 1 in 5 Americans during COVID-19 peak

Key lesson
Company failed due to rapid expansion and poor business strategy – not the AI itself

11xAI
Sales

What it does
AI sales development representative "Alice" researches accounts, writes e-mails and books meetings

Impact
$2M annual recurring revenue (ARR) one year after its founding

Key lesson
AI scales repetitive outreach, but human oversight remains critical for nuanced enterprise deals

Ema
Enterprise

What it does
"Universal AI employee" handles sales, HR, customer service and admin across business functions

Impact
$25M raised from investors. Clients include TrueLayer and Moneyview.

Key lesson
One generalist AI agent can replace multiple specialized tools, cutting costs for small firms like start-ups

Case studies show AI agent impact across industries (2/2)

Norm AI
Compliance

What it does
AI agents automate KYC/KYB checks, due diligence and compliance flagging

Impact
Saves weeks of manual review in financial services.

Key lesson
In highly regulated industries, AI augments human decisions rather than replacing them entirely

Composabl
Industrial

What it does
AI agents control and adjust industrial equipment in real time – no human intervention needed

Impact
Continuous process optimization for efficiency and output

Key lesson
AI agents move beyond pure office work into physical operations like manufacturing

Tech Leaders
Platforms

What they do
Microsoft (AutoGen), Google (Vertex AI), Salesforce (Agentforce) and AWS (Bedrock) launched agent platforms

Impact
Enterprise-grade agent building with no-code tools, RAG, etc.

Key lesson
AI agents become part of platform infrastructure – not just standalone products

Takeaways

Dynamic workflows replace rigid processes
AI agents enable real-time adaptive workflows with continuous learning and predictive capabilities across departments

Full automation transforms key areas
… like data processing, translation, quality control, customer service, predictive maintenance and SC optimization

Human-AI collaboration spans four maturity levels
From AI agent as assistant to autonomous organizations – matching the right framework to the task is critical

Uniquely human skills remain irreplaceable
Complex negotiations, strategic leadership and creative innovation – among others – resist automation

Workplace readiness requires a structured approach
… including building data infrastructure, upskilling workforce and redesigning workflows

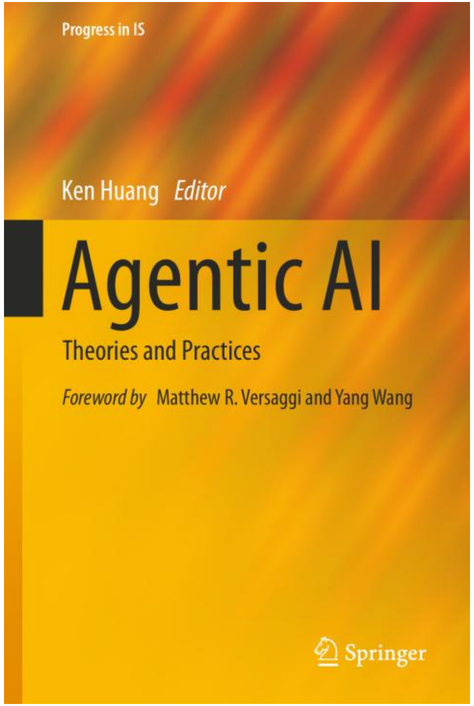
Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes

References



Book by Ken Huang (2025) with the title
“Agentic AI: Theories and Practices”

This lecture focuses particularly on:
- Chapter 5: AI Agents and Business Workflow

Link to HTW library: https://sisis.rz.htw-berlin.de/search?bvnr=BV050372763

– Please re-read the corresponding parts –

Quiz time!

Go to slido.com

and join with **# 1899**

Thank you for your attention

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